Data Analyst in Power BI with AI

A basic explanation of the six core topics

1 Excel and Data Basics
2 Power BI Desktop Essentials
3 Data Modelling
4 DAX Fundamentals
5 Visuals and Dashboards
6 AI Features and Publishing

Overview

A data analyst turns raw data into decisions. The typical workflow is: **collect and clean** data, **model** it, **calculate** useful numbers, **visualise** them, and **share** the result. This guide follows that path in six steps.

#	Topic	What you learn	Key tools
1	Excel and Data Basics	Understand and prepare data	Excel, Pivot Tables, XLOOKUP
2	Power BI Desktop Essentials	Connect and transform data	Power BI Desktop, Power Query
3	Data Modelling	Organise tables so they work together	Star schema, relationships
4	DAX Fundamentals	Create calculations and KPIs	Measures, CALCULATE
5	Visuals and Dashboards	Present insights clearly	Charts, slicers, drill-through
6	AI Features and Publishing	Use AI and share reports	Copilot, Q&A, Power BI Service

Tip: Learn the topics in order. Each one builds on the previous one - clean data (1-2) makes a good model (3), which makes DAX (4) simple, which makes visuals (5) meaningful.

1 Excel and Data Basics

Data types, cleaning, pivot tables, lookup formulas

Data types

Every column holds one kind of data. Using the correct type keeps calculations and sorting accurate.

Type	Example	Used for
Text	Laptop, Delhi	Names, categories, IDs
Number (whole / decimal)	25, 199.50	Quantity, price, sales
Date / Time	20-Oct-2026	Trends over time
Boolean	TRUE / FALSE	Yes/No flags

Data cleaning

Real-world data is messy. Cleaning means fixing it before analysis:

- **Remove duplicates** - Data tab > Remove Duplicates.
- **Handle blanks** - fill, replace or remove empty cells.
- **Fix spelling and spaces** - use TRIM(), PROPER(), UPPER(), CLEAN().
- **Correct data types** - dates stored as text, numbers stored as text.
- **Split / combine columns** - Text to Columns, or & and TEXTJOIN().
- **Keep data in a table** - press Ctrl+T so formulas and pivots expand automatically.

Pivot Tables

A pivot table summarises thousands of rows in seconds without writing formulas. Select your table, then Insert > PivotTable and drag fields into four areas:

- **Rows** - categories to group by (Region, Product).
- **Columns** - a second grouping (Year, Quarter).
- **Values** - the numbers to summarise (Sum of Sales, Count of Orders).
- **Filters** - limit the whole report (Country = India).

Lookup formulas

Lookups pull matching information from another table, for example a product price from a price list.

```
=XLOOKUP(A2, Products[ID], Products[Price], "Not found")
=VLOOKUP(A2, Products!A:C, 3, FALSE)
=INDEX(Products[Price], MATCH(A2, Products[ID], 0))
```

Tip: Prefer XLOOKUP over VLOOKUP: it looks left or right, defaults to exact match and handles missing values.

2 Power BI Desktop Essentials

Interface, data sources, Power Query transformations

What is Power BI Desktop?

Power BI Desktop is a free Windows application used to connect to data, transform it, build a model and design reports.

The interface

Area	Purpose
Report view	Canvas where you build charts and dashboards
Table view	Inspect the data loaded into the model
Model view	See tables and draw relationships
Fields pane	Lists tables, columns and measures to drag onto visuals
Visualizations / Filters panes	Choose chart types, format them and apply filters
Ribbon	Home, Insert, Modeling and View tools

Data sources

Use **Home > Get Data** to connect to almost anything:

- **Files** - Excel, CSV, JSON, PDF, folders.
- **Databases** - SQL Server, MySQL, PostgreSQL, Oracle.
- **Online services** - SharePoint, Google Analytics, Salesforce.
- **Web** - tables from a web page.

Import vs DirectQuery

Import copies data into Power BI (fast, most common). **DirectQuery** leaves data in the source and queries it live (always current, but slower).

Power Query transformations

Power Query (Transform Data) is where you clean and shape data. Every step is recorded in **Applied Steps** and replays automatically when the data refreshes.

- Promote the first row to headers; set correct data types.
- Remove columns, rows, duplicates and errors.
- Filter rows, split or merge columns, replace values.
- **Merge Queries** (join tables) and **Append Queries** (stack tables).
- **Group By**, **Pivot** / **Unpivot** columns, add a **Custom Column**.

Tip: Do heavy cleaning in Power Query, not in the report. Clean once, reuse on every refresh.

3 Data Modelling

Star schema, relationships, calculated columns

Why model data?

A good model connects separate tables so that filters flow correctly and calculations stay fast and accurate.

Star schema

The recommended design has one central **fact table** surrounded by **dimension tables**, like a star.

Table type	Contains	Example
Fact	Events / transactions, numeric values, keys	Sales (OrderID, DateKey, ProductKey, Amount)
Dimension	Descriptive attributes used to filter and group	Product, Customer, Date, Region

```
          Date          Product
            \            /
             \          /
              [  Sales  ]      <-- fact table (center)
             /          \
            /            \
       Customer          Region
```

Relationships

A relationship links a column in one table to a column in another (for example Sales[ProductKey] to Product[ProductKey]).

- **One-to-many (1:*)** - the standard type: one product, many sales.
- **Many-to-many** - use sparingly; it can cause confusing results.
- **Cross-filter direction** - keep it *single* (dimension filters fact) unless you have a clear reason.
- **Active vs inactive** - only one active relationship between two tables; others are used via DAX.

Calculated columns

A calculated column adds a new column to a table using DAX. It is calculated row by row when data loads and stored in the model.

```
Profit = Sales[Revenue] - Sales[Cost]
Full Name = Customer[First] & " " & Customer[Last]
```

Tip: Use a calculated column for values you want to slice by (such as Age Group). Use a measure (next section) for totals and KPIs.

4 DAX Fundamentals

Measures, CALCULATE, time intelligence, KPIs

What is DAX?

DAX (Data Analysis Expressions) is the formula language of Power BI. It looks like Excel formulas but works on whole tables and columns.

Measures

A measure is a calculation evaluated on demand, responding to filters, slicers and visuals. It is not stored per row, so it keeps files small and results dynamic.

```
Total Sales = SUM(Sales[Amount])
Avg Order Value = DIVIDE([Total Sales], COUNTROWS(Sales))
```

Common functions

Group	Functions
Aggregation	SUM, AVERAGE, MIN, MAX, COUNT, DISTINCTCOUNT
Logical	IF, SWITCH, AND, OR
Safe maths	DIVIDE (avoids divide-by-zero errors)
Iterators	SUMX, AVERAGEX - calculate row by row, then aggregate

CALCULATE

CALCULATE is the most important DAX function. It evaluates an expression after **changing the filters**.

```
India Sales = CALCULATE([Total Sales], Customer[Country] = "India")
All Region Sales = CALCULATE([Total Sales], ALL(Region))
```

Time intelligence

Time functions compare periods. They require a proper **Date table** (continuous dates, marked as a date table) related to your fact table.

```
Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])
Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
YoY Growth % = DIVIDE([Total Sales] - [Sales LY], [Sales LY])
```

KPIs

KPIs measure performance against a goal. Combine measures to build them (keep them in a dedicated Measures table and reuse them):

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales])
```

5 Visuals and Dashboards

Charts, slicers, drill-through, storytelling

Choosing the right chart

Question you want to answer	Best visual
How does a value change over time?	Line chart, area chart
How do categories compare?	Bar / column chart
What share does each part hold?	Donut or pie (few categories), treemap
Is there a relationship between two numbers?	Scatter chart
What is the single headline number?	Card, KPI, gauge
Where do I need exact values?	Table, matrix
Where is it happening?	Map, filled map

Slicers and filters

- **Slicers** are on-page buttons or lists (Year, Region) that let viewers filter visuals.
- **Filter pane** filters at visual, page or whole-report level.
- **Edit interactions** controls which visuals react to a selection.
- **Sync slicers** applies one slicer across multiple pages.

Drill-down and drill-through

- **Drill-down** moves through a hierarchy inside a visual (Year > Quarter > Month).
- **Drill-through** sends the viewer to a detail page for the item they right-clicked, for example from a Region summary to a page showing that region's customers.

Dashboard storytelling

A dashboard should answer a question, not just display numbers.

- **Top-left first:** place the most important KPI cards at the top.
- **Less is more:** 5-8 visuals per page; remove clutter.
- **Consistent colours:** use colour to highlight, not decorate.
- **Clear titles:** write insight titles such as 'Sales grew 12% year on year'.
- **Logical flow:** overview first, then detail through drill-through pages.
- **Use bookmarks and buttons** for navigation and guided views.

Tip: Before publishing, ask: 'Can a new viewer understand the main message in 10 seconds?'

6 AI Features and Publishing

Copilot, Q&A visual, Power BI Service, sharing

AI features in Power BI

Feature	What it does
Copilot	A generative-AI assistant. Describe what you want in plain language and it can create report pages, suggest visuals, write DAX, summarise a report and answer questions about your data.
Q&A visual	Type a question such as 'total sales by region in 2026' and Power BI builds the chart automatically.
Smart narrative	Generates an automatic text summary of the visuals and trends.
Key influencers	Shows which factors most affect a result (for example, what drives churn).
Decomposition tree	Breaks a number down by different categories, step by step, to find root causes.
Anomaly detection / Forecast	Flags unusual points on a line chart and projects future values.

Tip: Copilot works best on a clean, well-named model. Always verify AI-generated DAX and results. Availability depends on your licence and capacity.

Power BI Service

Power BI Service is the online (cloud) platform at app.powerbi.com. After building a report in Desktop, click **Publish** to send it to a workspace.

- **Workspace** - a shared area where teams store reports, datasets and dashboards.
- **Semantic model (dataset)** - the data and model behind your reports.
- **Report** - multi-page, interactive visuals. **Dashboard** - single page of pinned tiles.
- **Scheduled refresh** - keeps data up to date automatically (a gateway is needed for on-premises sources).
- **Row-level security (RLS)** - limits what data each user can see.

Sharing

Method	Best for
Share a report link	Quick sharing with specific colleagues
Workspace roles (Admin, Member, Contributor, Viewer)	Team collaboration and controlled editing
Power BI App	Packaging reports for a wider audience
Subscribe / email	Regular snapshots delivered to inboxes
Export (PDF, PowerPoint, Excel)	Offline use or presentations
Embed / Publish to web	Websites or portals (public publish exposes data to everyone)

Most sharing features need a Power BI Pro / Premium Per User licence or a Fabric capacity.

Quick Recap and Next Steps

Step	Remember
1. Excel and Data Basics	Clean data, use correct types, summarise with pivots, match with XLOOKUP.
2. Power BI Desktop	Connect to sources and shape data in Power Query.
3. Data Modelling	Star schema with one-to-many relationships; columns for slicing.
4. DAX	Measures for calculations; CALCULATE changes filters; use a Date table.
5. Visuals	Right chart, simple layout, slicers, drill-through, clear story.
6. AI and Publishing	Copilot and Q&A speed up insight; publish, refresh and share securely.

Practice project idea

Download a free sales dataset (for example Superstore). Clean it in Power Query, build a star schema, write measures for Total Sales, Profit Margin and YoY Growth, design a 2-page dashboard with a drill-through, then publish it and test Copilot and the Q&A visual.

Suggested learning order

- Week 1: Excel, pivot tables and lookups.
- Week 2: Power BI Desktop and Power Query.
- Week 3: Data modelling and basic DAX.
- Week 4: Advanced DAX, visuals and storytelling.
- Week 5: Publishing, sharing, AI features and a final project.